

Name: _____

Exam Style Questions

Expanding Two Brackets



Corbettmaths

Equipment needed: Calculator, pen

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

Video 14



Answers and Video Solutions



1. Expand and simplify $(y + 3)(y + 5)$



$$y^2 + 5y + 3y + 15$$

$$y^2 + 8y + 15$$

$$\frac{y^2 + 8y + 15}{(2)}$$

2. Expand and simplify $(x + 5)(x - 1)$



$$x^2 - x + 5x - 5$$

$$x^2 + 4x - 5$$

$$\frac{x^2 + 4x - 5}{(2)}$$

3. Expand and simplify $(w - 2)(w - 7)$



$$w^2 - 7w - 2w + 14$$

$$w^2 - 9w + 14$$

$$\frac{w^2 - 9w + 14}{(2)}$$

4. Expand and simplify $(x - 10)(x + 3)$



$$x^2 + 3x - 10x - 30$$

$$x^2 - 7x - 30$$

$$\frac{x^2 - 7x - 30}{(2)}$$

5.

Expand and simplify

$(x - 4)(x + 6)$



$$x^2 + 6x - 4x - 24$$

$$x^2 + 2x - 24$$

$$x^2 + 2x - 24$$

(2)

6.

Expand and simplify

$(x - 3)(x + 3)$



$$x^2 + 3x - 3x - 9$$

$$x^2 - 9$$

$$x^2 - 9$$

(2)

7.

Expand and simplify

$(x - 7)^2$



$$(x - 7)(x - 7)$$

$$x^2 - 7x - 7x + 49$$

$$x^2 - 14x + 49$$

$$x^2 - 14x + 49$$

(2)

8.

Expand and simplify

$(2y + 1)(y + 3)$



$$2y^2 + 6y + y + 3$$

$$2y^2 + 7y + 3$$

$$2y^2 + 7y + 3$$

(2)

9. Expand and simplify $(4x + 1)(3x - 5)$



$$12x^2 - 20x + 3x - 5$$

$$12x^2 - 17x - 5$$

$$\underline{12x^2 - 17x - 5}$$

(2)

10. Expand and simplify $(3x - 2)(2x + 3)$



$$6x^2 + 9x - 4x - 6$$

$$6x^2 + 5x - 6$$

$$\underline{6x^2 + 5x - 6}$$

(2)

11. Expand and simplify $(5y - 1)(y - 2)$



$$5y^2 - 10y - y + 2$$

$$5y^2 - 11y + 2$$

$$\underline{5y^2 - 11y + 2}$$

(2)

12. Expand and simplify $(7x - 20)(9x - 10)$



$$63x^2 - 70x - 180x + 200$$

$$63x^2 - 250x + 200$$

$$\underline{63x^2 - 250x + 200}$$

(2)

13. Expand and simplify $(2x + 5)^2$



$$(2x + 5)(2x + 5)$$

$$4x^2 + 10x + 10x + 25$$

$$4x^2 + 20x + 25$$

$$\underline{4x^2 + 20x + 25}$$

(2)

14. Expand and simplify $(4x - 3)^2$



$$(4x - 3)(4x - 3)$$

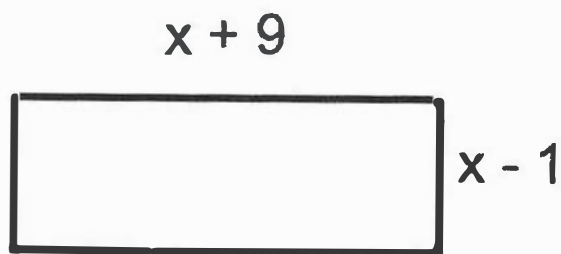
$$16x^2 - 12x - 12x + 9$$

$$16x^2 - 24x + 9$$

$$\underline{16x^2 - 24x + 9}$$

(2)

15. A rectangle is shown below.



The length of the rectangle is $x + 9$ cm.

The width of the rectangle is $x - 1$ cm.

Form an expression for the area of the rectangle.

$$(x + 9)(x - 1)$$

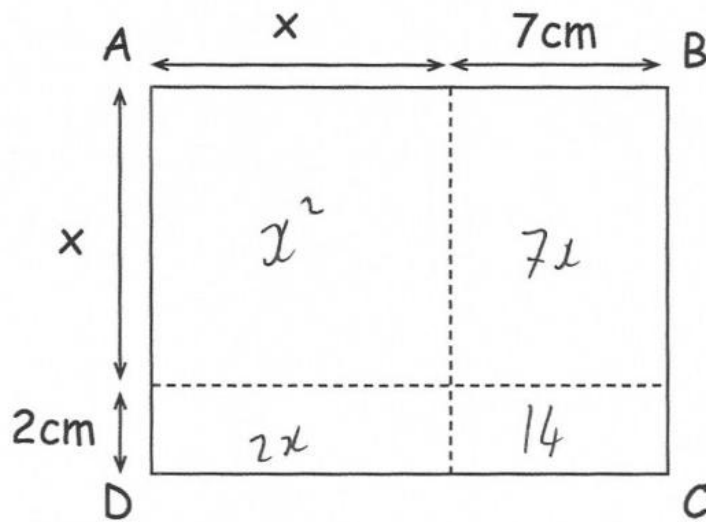
$$x^2 - x + 9x - 9$$

$$x^2 + 8x - 9$$

$$x^2 + 8x - 9$$

(3)

16. Rectangle ABCD is shown below.



The area rectangle ABCD is 230cm^2

Show that $x^2 + 9x = 216$ $(x+7)(x+2)$

$$x^2 + 7x + 2x + 14 = 230$$

$$x^2 + 9x + 14 = 230$$

$$x^2 + 9x = 216$$

(4)

17. Expand and simplify $(3 + g)(5 - g)$



$$15 - 3g + 5g - g^2$$

$$15 + 2g - g^2$$

$$\frac{15 + 2g - g^2}{(2)}$$

18. Micah correctly expands and simplifies $(2x + 11)(x - 3)$



Circle the term that is part of his answer

$8x$

$-5x$

$5x$

$-8x$

$$2x^2 - 6x + 11x - 33$$

$$2x^2 + 5x - 33$$

(1)

19. Expand and simplify $(y^2 + y)(y + 3)$



$$y^3 + 3y^2 + y^2 + 3y$$

$$y^3 + 4y^2 + 3y$$

$$y^3 + 4y^2 + 3y$$

(2)

20. Expand and simplify $(w + 3)(w + 4) + (w + 2)(w + 7)$



$$w^2 + 4w + 3w + 12$$

$$w^2 + 7w + 12$$

$$w^2 + 7w + 2w + 14$$

$$w^2 + 9w + 14$$

$$2w^2 + 16w + 26$$

$$2w^2 + 16w + 26$$

(3)

21. Expand and simplify $(3y - 5)(y + 4) - (y - 3)(y - 5)$



$$3y^2 + 12y - 5y - 20 \quad y^2 - 5y - 3y + 15$$
$$3y^2 + 7y - 20 - (y^2 - 8y + 15)$$

$$\frac{2y^2 + 15y - 35}{(3)}$$

22. Expand and simplify $(4y^2 + 5)(2y + 1) - 3y(y^2 - 6)$



$$8y^3 + 4y^2 + 10y + 5 - 3y^3 + 18y$$

$$5y^3 + 4y^2 + 28y + 5$$

$$\frac{5y^3 + 4y^2 + 28y + 5}{(4)}$$

23. $(x + c)(x + d) \equiv x^2 + px + 72$



c is twice the value of d

Find two possible values of p

$$x^2 + dx + cx + cd \equiv x^2 + px + 72$$

$$cd = 72$$

$$c = 12 \quad d = 6$$

$$c = -12 \quad d = -6$$

$$(x + 12)(x + 6) = x^2 + 6x + 12x + 72$$

$$x^2 + 18x + 72$$

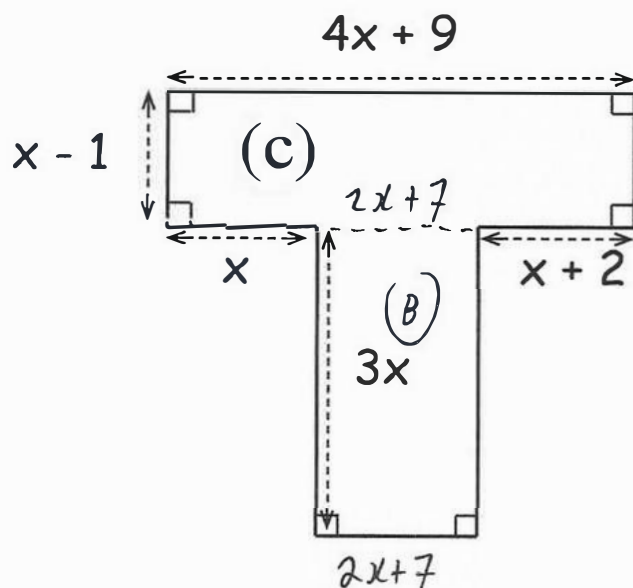
$$(x - 12)(x - 6) = x^2 - 6x - 12x + 72$$

$$= x^2 - 18x + 72$$

$$p = 18 \text{ or } p = -18$$

(3)

24.



The area of the shape shown above is A .

Show that $A = 10x^2 + 26x - 9$

$$\begin{aligned} (C) \quad (4x + 9)(x - 1) &= 4x^2 - 4x + 9x - 9 \\ &= 4x^2 + 5x - 9 \end{aligned}$$

$$(B) \quad 3x(2x + 7) = 6x^2 + 21x$$

$$(C) + (B) = 10x^2 + 26x - 9$$

Total Area = A

$$A = 10x^2 + 26x - 9$$

(4)